From Control to Offline Reinforcement Learning

Learning Better Decisions from Logged Data

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Predictions Are Not Decisions

Supervised learning predicts	RL chooses
Truck arrival time	Which truck to unload next
Unloading duration	Which pit to assign
Grain quality	Which silo to route into
Train delay	How to schedule train loading

Main takeaway

Supervised learning predicts what happens next. Reinforcement learning chooses what should happen next.

Actions Change the Future

Choosing a truck, pit, route, or silo changes future queues, storage, blending options, and train schedules.

Main takeaway

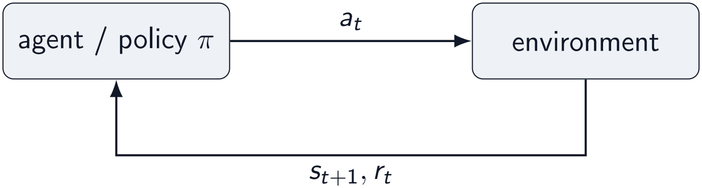
In RL, the model does not just observe the data distribution. It helps create the next data distribution.

States, Actions, and Policies

Object	Grain-terminal example
state s_t	current queue, silo inventory, grain quality, train progress
action a_t	unload truck, choose pit, route conveyor, assign silo, fill car
policy $\pi(a \mid s)$	rule or model that chooses actions from states
next state s_{t+1}	the facility after the action has changed it

A decision problem starts with states, actions, and a policy that maps situations to decisions.

Agent-Environment Loop



- **Agent:** chooses the next action from the current state.
- **Environment:** the facility or simulation that returns the next state and reward.
- **Key contrast:** supervised learning predicts from fixed data; RL actions create the next data.

Imitation Learning: Turn Decisions Into Supervised Learning

First idea: collect observed behavior and train a policy to predict what the operator would do.

Historical demonstrations come from a behavior policy π_b, e.g. human terminal operators:

$$\mathcal{D} = \{(s_t, a_t)\}, \qquad a_t \sim \pi_b(a \mid s_t)$$

Train π_θ as a supervised action predictor:

Continuous actions
steering angle, conveyor speed

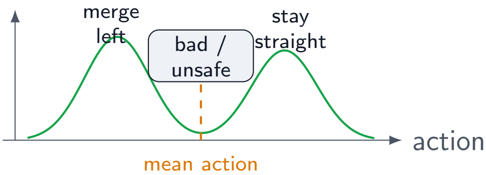
$$\min_\theta \; \mathbb{E}_\mathcal{D}\left[\|a - \pi_\theta(s)\|^2\right]$$

Discrete actions
route choice, silo assignment

$$\min_\theta \; -\mathbb{E}_\mathcal{D}\left[\log \pi_\theta(a \mid s)\right]$$

Attractive because it is fully offline, simple to train, and does not require defining a reward.

Pitfall 1: Averaging Demonstrations Can Fail



Consider imitation data for self-driving cars:

- in similar observed states, driver 1 stays in lane
- driver 2 changes lanes
- squared-error regression learns the conditional mean action

Problem: the mean action can lie between valid modes: a poor command that no expert demonstrated.

Pitfall 2: Errors Accumulate Over Time

t: small error

↓

$t+1$: shifted state

↓

$t+2$: bigger error

↓

$t+3$: unfamiliar state

A policy trained only on expert states may not know how to recover from states caused by its own mistakes.

- $\rightarrow$ Distribution shift: deployment samples from the learned policy's state distribution, not the expert's.

Problem: one-step prediction errors can look small while long rollouts become unstable.

Pitfall 3: Imitation Ignores Outcomes

Behavior cloning asks only:

"What action was taken in this situation?"

It does not ask:

"Which action led to the best long-term outcome?"

Every logged decision becomes a training example. The quality of the outcome does not change how strongly it is learned.

Problem: behavior cloning reproduces past decisions, but it cannot prefer decisions with better long-term consequences.

Rewards Measure the Consequences of Decisions

A reward turns delayed product outcomes into feedback for decisions:

$$r_t = r(s_t, a_t, s_{t+1})$$

For grain loading, rewards can combine:

- train loading progress and throughput
- contract quality satisfaction
- penalties for rehandling, congestion, and delay

Rewards tell the learner what outcomes matter, not just what actions were taken.

Long-Term Consequences

A good decision is one that leads to *high future reward*, not just a high immediate reward.

$$\text{Future Reward} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots$$

where $0 < \gamma < 1$ discounts delayed rewards.

A value function predicts this quantity before making a decision:

$$V(s) \;\approx\; \text{future reward from this state}$$

$$Q(s, a) \;\approx\; \text{future reward after taking action } a$$

Main takeaway

V and Q quantify how good it is to be in a state and to take an action. They are central objects in RL.

Using Advantage to Weight Data

Advantage scores a logged action relative to what is typical in the same state:

$$A(s, a) = Q(s, a) - V(s)$$

- positive advantage: this logged action was better than usual
- negative advantage: this logged action was worse than usual

Convert advantage into a training weight:

$$\begin{aligned} \text{advantage weight:} \quad & w(s,a) = \exp(\beta A(s,a)), \qquad \beta > 0 \\ \text{advantage-weighted BC:} \quad & \min_\theta \sum_{(s_i,a_i)\in\mathcal{D}} w(s_i,a_i)\,\left\|\pi_\theta(s_i) - a_i\right\|^2 \end{aligned}$$

Simplest offline RL idea

Count "good" data points multiple times, and "bad" data points only as a fraction.

(For now, assume we have the values. We will discuss how to compute them later.)

From Decisions to MDPs

A Markov Decision Process adds dynamics and rewards:

$$(\mathcal{S}, \mathcal{A}, P, r, \gamma)$$

- $\mathcal{S}$: possible states
- $\mathcal{A}$: possible actions
- $P(s' \mid s, a)$: transition
- $r(s, a)$: reward
- $\gamma \in [0, 1)$: discount

The key modeling assumption is that s_t contains enough information to make the next decision.

Bellman Thinking

$$Q(s, a) \approx r(s, a) + \gamma V(s')$$

The value of an action is immediate reward plus the value of where it takes us.

Bellman recursion is the RL version of cost-to-go recursion in optimal control.

From Value to Action: Learn $Q(s, a)$

If we know $Q(s, a)$, we can compare actions in the same state.

choose the action with the highest predicted long-term value

This is why many RL algorithms learn an action-value function.

Q-Learning Update

A one-step target for Q:

$$Q(s_t, a_t) \leftarrow r_t + \gamma \max_{a'} Q(s_{t+1}, a')$$

Immediate reward plus the best predicted value at the next state.

Greedy Policy Improvement

Once we have Q, improve the policy by acting greedily:

$$\pi_{\mathrm{new}}(s) = \arg\max_a Q(s, a)$$

This is the basic RL loop:

estimate values $\rightarrow$ choose better actions.

Why Q-Learning Is Dangerous Offline

Offline data only covers some actions.

$$\max_a Q(s, a)$$

The max may select an action the dataset barely supports, where Q is just a guess.

Main takeaway

The same maximization that improves policies online can exploit value errors offline.

Control $\leftrightarrow$ RL Dictionary

Control theory	**Reinforcement learning**
State x_t	State s_t
Control u_t	Action a_t
Dynamics f	Transition $P(s' \mid s, a)$
Cost $c(x, u)$	Negative reward $-r(s, a)$
Controller	Policy $\pi(a \mid s)$
Cost-to-go $J(x)$	Value function $V(s)$
Dynamic programming / HJB	Bellman equations
MPC	Model-based RL / planning

PID as a Simple Policy Class

A PID controller chooses an action from tracking error:

$$a_t = K_P e_t + K_I \sum_{\tau \le t} e_\tau + K_D (e_t - e_{t-1})$$

Define the state to include the PID features:

$$s_t = (y_t,\ e_t,\ I_t,\ D_t)\,, \quad I_t = \sum_{\tau \le t} e_\tau\,, \quad D_t = e_t - e_{t-1}.$$

Then PID is a deterministic policy:

$$a_t = \pi_K(s_t) = K_P e_t + K_I I_t + K_D D_t, \qquad K = (K_P, K_I, K_D).$$

The environment remains the transition:

$$s_{t+1} \sim p(s_{t+1} \mid s_t, a_t).$$

Since K parameterizes a policy, RL can tune K by optimizing return.

PID and RL: Different Strengths

Classical / PID control	Reinforcement learning
Simple and interpretable	Optimizes delayed objectives
Strong local feedback	Handles sequential tradeoffs
Easier to validate	Learns from data and simulation
Needs hand-designed structure	Can adapt to nonlinear operations
Usually optimizes a proxy	Can optimize the KPI directly

Main takeaway

Classical control is often the right starting point. RL becomes attractive when the best action depends on long-horizon operational consequences.

Why RL Should Improve Long Term

- It can optimize delayed outcomes, not only immediate tracking error.
- It can use simulators to evaluate actions not yet tried in production.
- It can learn terminal-specific interactions among queues, silos, conveyors, and trains.
- It can update as operating conditions and objectives change.

For grain loading, the best local move may not be the best move for throughput, quality, rehandling, and train delay.

Online RL Loop

Online RL learns by interacting while training.

Three Common RL Families

Family	Basic idea
Value-based RL	Learn $Q(s, a)$, act greedily
Policy gradient	Directly improve $\pi(a \mid s)$
Actor-critic	Learn both policy and value

Why Online RL Is Hard in Products

- Unsafe or costly exploration
- Customer-facing experiments
- Long feedback delays
- Sparse or noisy rewards
- Operational constraints and compliance

When exploration is expensive, risky, or product-constrained, logged data becomes the natural starting point.

Model-Based RL: Learn the Dynamics, Then Plan

$$\hat{s}_{t+1} = \hat{f}_\theta(s_t, a_t)$$

$$a^\star_{0:H} = \arg\max_{a_{0:H}} \sum_{t=0}^{H} \gamma^t \hat{r}(s_t, a_t)$$

Model-based RL often looks like MPC with a learned dynamics model.

Risks: model bias, uncertainty, and compounding rollout error.

Offline RL Setting

Fixed dataset:

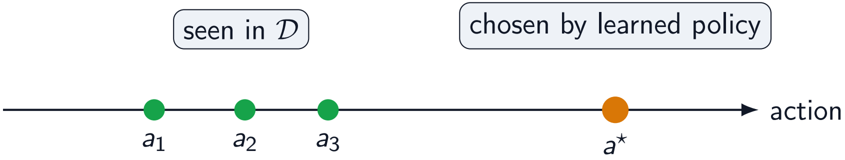
$$\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}$$

Behavior policy:

$$\pi_b(a \mid s)$$

Goal: learn $\pi(a \mid s)$ without online interaction during training. Logged data source: terminal operator logs. Action space: truck selection, pit assignment, conveyor route, silo, and loading schedule.

Support Mismatch



The policy may choose an unsupported action where the value estimate is unreliable.

Why Naive Q-Learning Fails Offline

Dangerous backup:

$$\max_a Q(s, a)$$

The max can select actions that were rarely or never observed.

Then the learned policy may exploit errors in Q outside the data distribution.

The Offline RL Thesis

Offline RL is mostly about preventing the policy from exploiting value errors outside the data distribution.

The rest of the talk builds a ladder:

$$\text{BC} \to \text{AWR} \to \text{IQL} \to \text{IDQL}$$

Recall: Behavior Cloning Is the Offline Baseline

$$\mathcal{L}_{\mathrm{BC}}(\theta) = -\mathbb{E}_{(s,a)\sim\mathcal{D}}\left[\log \pi_\theta(a \mid s)\right]$$

Learn to imitate the actions in the offline dataset.

It is stable and useful, but it does not use rewards or long-term outcomes.

Offline RL starts from BC and asks how to improve it without leaving the data support.

Advantage-Weighted Regression

$$\mathcal{L}_{\mathrm{AWR}}(\theta) = -\mathbb{E}_{(s,a)\sim\mathcal{D}}\left[w(s,a)\log \pi_\theta(a \mid s)\right]$$

$$w(s,a) \propto \exp\left(\frac{A(s,a)}{\lambda}\right), \qquad A(s,a) = Q(s,a) - V(s)$$

BC vs. AWR Weighting

High-advantage actions are cloned more strongly; low-advantage actions are downweighted.

AWR Is the First RL Step

AWR is the smallest conceptual step from imitation learning to reinforcement learning.

It still looks like supervised learning, but the labels are now weighted by long-term value.

If logs contain good and bad decisions, AWR gives a way to imitate the better ones more.

IQL Critic Learning

Avoid explicit maximization over unseen actions:

$$\text{avoid: } \max_a Q(s, a)$$

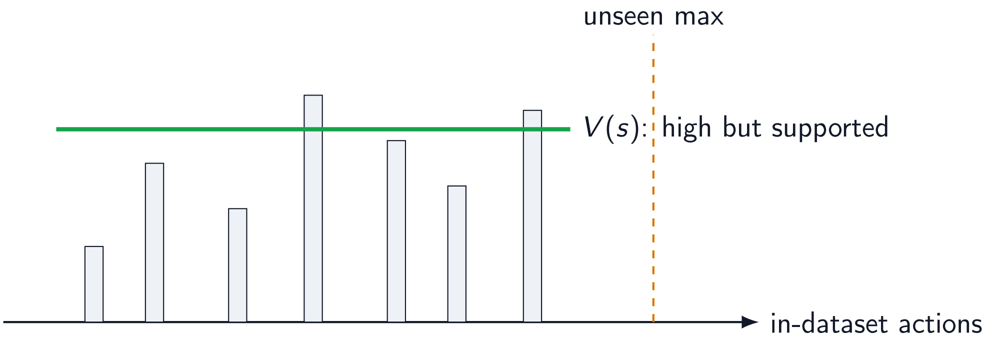
Dataset transition update:

$$Q(s, a) \leftarrow r + \gamma V(s')$$

Learn V as an upper expectile of in-dataset Q values:

$$\mathcal{L}_V = \mathbb{E}_{(s,a)\sim\mathcal{D}}\left[\left|\tau - 1_{Q(s,a)-V(s)<0}\right|(Q(s, a) - V(s))^2\right], \quad \tau > 0.5$$

In-Sample Value Learning



IQL estimates a high value from in-dataset actions instead of maximizing over all actions.

IQL Policy Extraction

$$\mathcal{L}_\pi = -\mathbb{E}_{(s,a)\sim\mathcal{D}}\left[\exp\left(\frac{Q(s,a) - V(s)}{\beta}\right)\log \pi_\theta(a \mid s)\right]$$

IQL uses advantage-weighted policy extraction after in-sample value learning.

IDQL: When the Action Distribution Is Multimodal

A simple Gaussian policy may average two good actions into a bad one.

Diffusion Policies: Generate Actions by Denoising

$\pi_\psi(a_{k-1} \mid a_k, s)$

A diffusion policy represents $\pi_\psi(a \mid s)$ as an iterative conditional generation process [1].

Instead of predicting one mean action, it can sample diverse plausible actions.

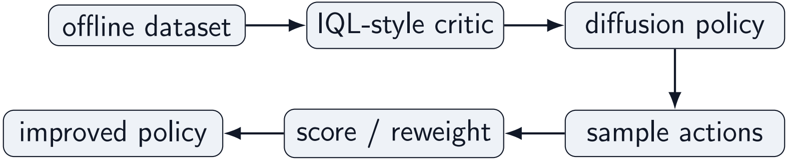
Why Diffusion Policies Fit Offline RL

- They can model multimodal action distributions.
- They stay anchored to logged behavior through behavior-model training.
- They expose multiple candidate actions for critic scoring.
- They are useful when action spaces are continuous, structured, or high-dimensional.

Main takeaway

Diffusion policies separate representation from improvement: sample plausible actions first, then use value estimates to prefer better ones.

IDQL Pipeline



IDQL combines an IQL-style critic with diffusion-based policy extraction [2].

Main takeaway

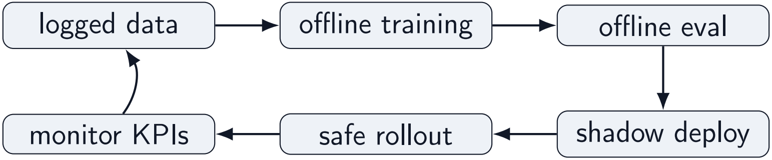
IQL tells us which actions are valuable; IDQL gives us a richer way to represent and sample them.

What Matters in Real Offline RL Systems?

1. **State quality:** is the decision approximately Markovian?
2. **Action support:** does $\mathcal{D}$ cover actions we want π to take?
3. **Reward design:** does reward reflect throughput, quality, delay, rehandling, and train-loading penalties?
4. **Counterfactual risk:** are we evaluating rare or unseen actions?
5. **Evaluation:** offline evaluation, simulation, shadow deployment, or cautious rollout?
6. **Policy constraints:** how far may π move from π_b?

Safe deployment strategy: shadow recommendations, operator approval, and constrained rollout.

Product Architecture Sketch



Closing Thesis

RL is data-driven optimal control.

Offline RL is controlled policy improvement from logged data.

The ladder:

$$\text{control} \rightarrow \text{RL} \rightarrow \text{offline RL} \rightarrow \text{BC} \rightarrow \text{AWR} \rightarrow \text{IQL} \rightarrow \text{IDQL}$$

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